Instrumentation Lab Session 4 LabVIEW Programming

Behind the 'Soft Front Panel' Instruments provided with ELVIS there is a powerful Graphical Programming Language which we can use to make our own instrument controls and displays. In this lab, you'll write a LabVIEW program to measure the gain versus frequency characteristics of the pre-amp you built in the previous lab session.

1. Objective

You have already carried out this measurement using the Bode Analyser. Here we'll see how we can write our own program in LabVIEW to achieve the same result. In essence, you will develop a bespoke measurement instrument using a combination of hardware and software. The advantage of this approach over using standard off-the-shelf equipment is that you can design and build your instrument to suit your specific needs instead of having to "make do" with the functions that manufacturers have decided to include in the instruments they sell. In the following exercise you will write a program that will drive the ELVIS hardware resources directly, instead of using the software instruments available in the NI ELVISmx Instrument Launcher as you have done so far.

2. Background Info

LabVIEW programs are called "virtual instruments", or VIs, because their appearance and operation imitate physical instruments, such as oscilloscopes and multimeters. LabVIEW boasts an extensive set of tools for acquiring, analyzing, displaying, and storing data, as well as tools to help you troubleshoot code you write.
In LabVIEW, you build a **user interface**, or **Front Panel**, with controls and indicators. Controls are knobs, push buttons, dials, etc. Indicators are graphs, LEDs, and other output displays. After you build the user interface, you add code using VIs and structures to control the front panel objects on the **Block Diagram**.
LabVIEW can be used to communicate with hardware such as data acquisition, vision, and motion control devices, as well as GPIB, PXI, VXI, RS232, and RS485 instruments.

3. Hardware Connections

The following hardware connections will be used throughout this script

- On the prototyping board connect FGEN to Pre Amp input
- Make the Pre Amp input available at CH0 (the BNC on the left-hand side of the ELVIS)
- Make the Pre Amp output available at CH1
- Switch on prototyping board

4. Starting LabVIEW

- Launch LabVIEW by clicking: Start >> Programs >> National Instruments >> LabVIEW 8.5 >> LabVIEW
- In the **Getting Started** window select **Blank VI**.
- You will be presented with two windows: the **Front Panel** and the **Block Diagram**.

Note: If the Front Panel is not visible, you can display it by selecting **Window»Show Front Panel**. You also can switch between the Front Panel window and Block Diagram window at any time by pressing the <Ctrl-E> keys.

5. Adding Controls and Indicators to the Front Panel

Controls on the Front Panel simulate the input mechanisms on a physical instrument and supply data to the Block Diagram of the VI.
Complete the following steps to add controls to the Front Panel.
Tip: Throughout these exercises, you can undo recent edits by selecting **Edit»Undo** or pressing the <Ctrl-Z> keys.

Step 1

- If the **Controls** palette shown in Figure 5-1 is not visible on the Front Panel, select **View » Controls Palette**.

If you are a new LabVIEW user, the **Controls** palette opens with the **Express** sub-palette visible by default. If you do not see the **Express** sub-palette, click **Express** on the **Controls** palette to display the **Express** sub-palette.
Tip: You can right-click any blank space on the Front Panel to display a temporary version of the **Controls** palettes. The **Controls** palette appears with a thumbtack icon in the upper left corner. Click the thumbtack to pin the palette so it is no longer temporary.

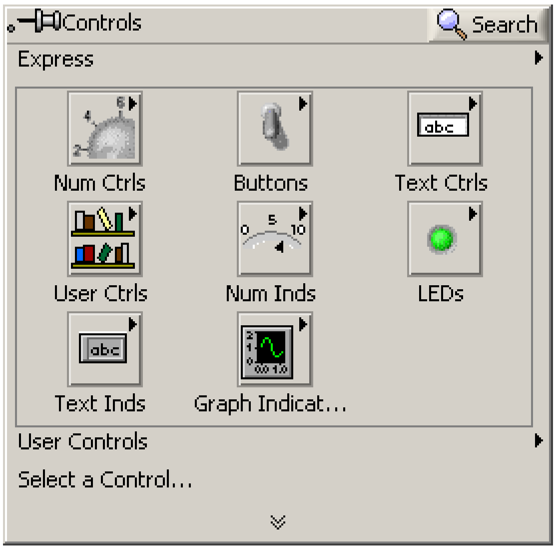

Figure 5-1, Controls Palette

Step 2

- Move the cursor over the icons on the **Express** sub-palette to locate the **Numeric Controls** palette.

When you move the cursor over icons on the **Controls** palette, the name of the sub-palette, control, or indicator appears in a tip strip below the icon.

Step 3

- Click the **Numeric Controls** icon to display the **Numeric Controls** palette.

Step 4

- On the **Numeric Controls** palette, click the **Numeric Control** item to attach the control to the cursor, and then place it on the Front Panel. Click again to drop it. As soon as you drop the control on the Front Panel, its label is highlighted and ready for editing. Change it from "Numeric" to "FGEN Start Frequency [Hz]".

Tip: You can change any label at any time by double clicking on them.

Step 5

- Repeat step 4 and add another **Numeric Control** to the front panel. Label it "FGEN Amplitude p-p [V]".

The two elements you have just added to the Front Panel are <u>**controls** i.e. elements that will be used to pass data to the program.</u>
Now we'll add some <u>**indicators** to the front panel, i.e. elements that are used to display data generated by the program.</u>

Step 6

- Repeat step 4 and add another **Numeric Control** to the front panel. Label it "Measured Gain".
- Right click on it and select **Change to Indicator** (you'll notice that the control arrows will disappear and the element will be greyed out indicating that the user cannot interact with it. It is now an **indicator**.)

Step 7

- Repeat step 6 and add another **Indicator** to the front panel. Label it "FGEN Current Frequency [Hz]".

Step 8

- Close the Numeric Control sub-palette and open the **Graph Indicators** sub-palette. Then click on **Graph** and drop it on the Front Panel.
- Change the Graph's label from "Waveform Graph" to "ELVIS/Channel 0"; change the Graph's X-axes label from "Time" to "Time [s]" and the Y-axes label from "Amplitude" to "Amplitude [V]".

Step 9

- Repeat step 8 and add another Graph indicator to the front panel. Label it "ELVIS/Channel 1" and change X and Y labels as in step 8.

Step 10

- Resize and reposition the various controls and indicators on the Front Panel so that you'll end up with the Front Panel looking like Figure 5-3.
- To **move** an element: click on the element you want to move and drag it to the desired location. Then drop it.
- To resize an element: move the mouse cursor on the edge of the element you want to resize; markers appear around it (see Figure 5-2). Click on a marker and drag to resize the element.

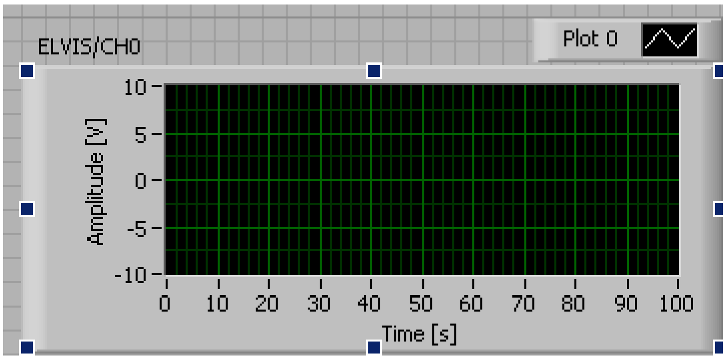

Figure 5-2

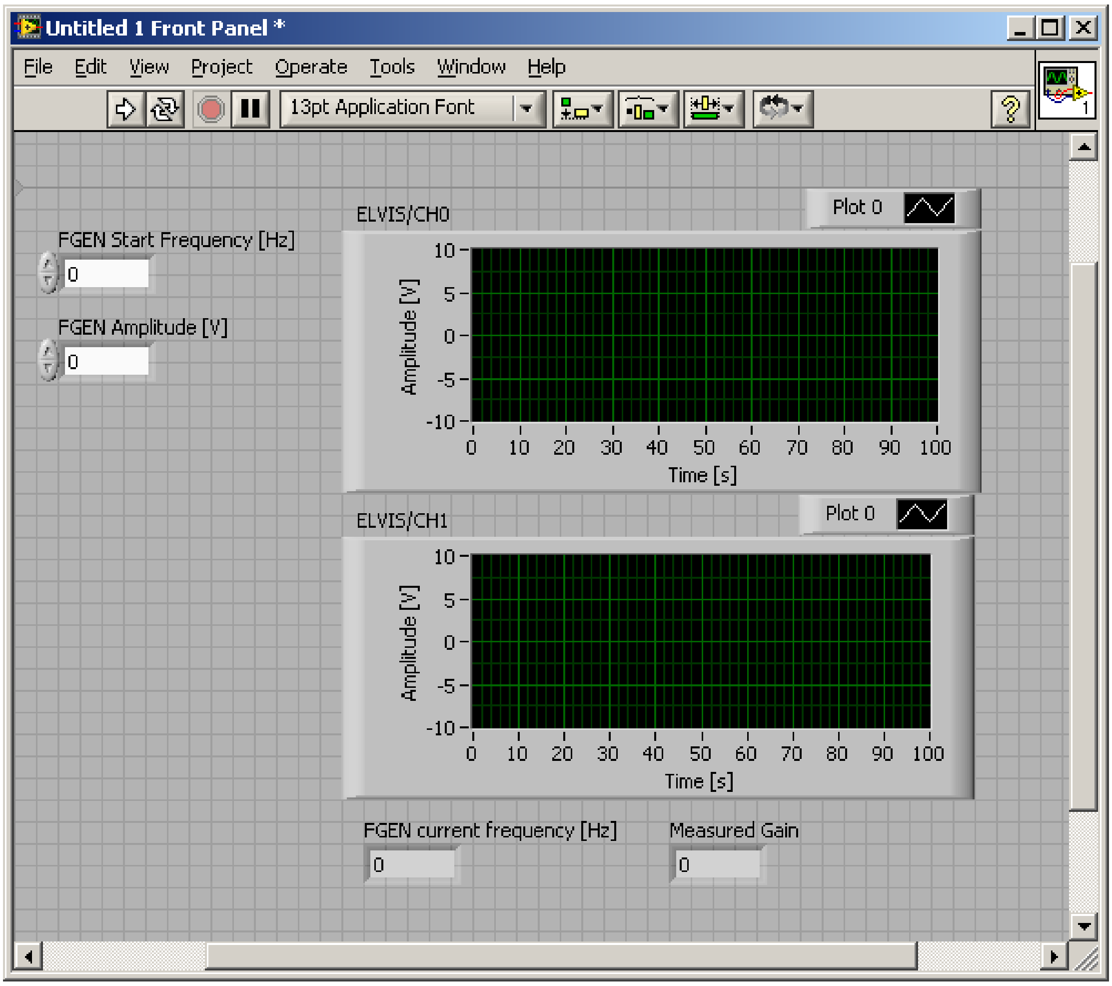

Figure 5-3

Step 11

➢ Select **File»Save As** and save the VI as “GainPlot.vi” in an easily accessible location.

6. Adding code to the Block Diagram

So far we have placed controls and indicators on the front panel but we have not done any programming.

➢ Try running the VI (press the button with the arrow on the top left corner of the Front Panel). Nothing happens.

Go to the Block Diagram. At this stage, your Block Diagram should look similar to Figure 6-1.

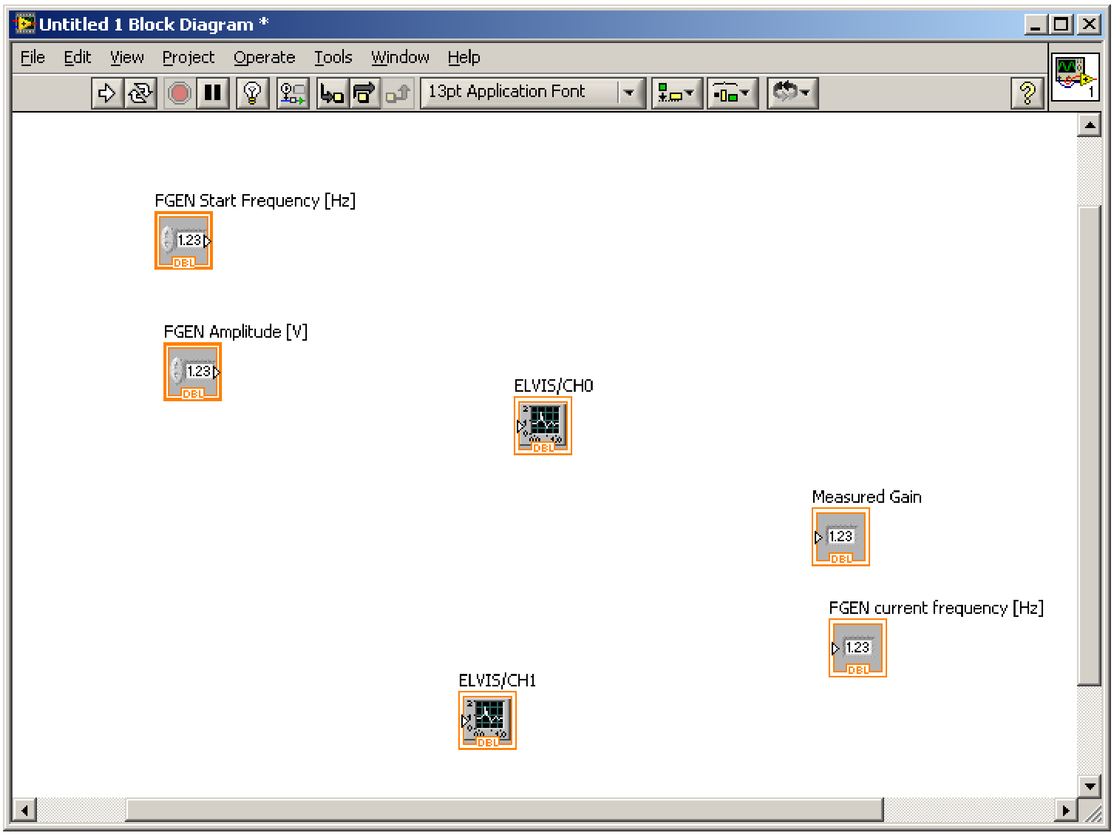

Figure 6-1

You will notice that all the control and indicators you placed on the Front Panel are available as **terminals** on the Block Diagram for you to use in you program.

Tip: you can click and drag any of the elements on the Block Diagram to reposition them.

Step 1

➢ If the **Function** palette shown in Figure 6-2 is not visible on the Block Diagram, select **View»Functions Palette** to display it.

Tip: You can right-click any blank space on the Block Diagram to display a temporary version of the **Functions** palette. The **Functions** palette appears with a thumbtack icon in the upper left corner. Click the thumbtack to pin the palette so it is no longer temporary.

If you are a new LabVIEW user, the **Functions** palette opens with the **Express** sub-palette visible by default. If you do not see the **Express** sub-palette, click **Express** on the **Functions** palette to display the **Express** sub-palette.

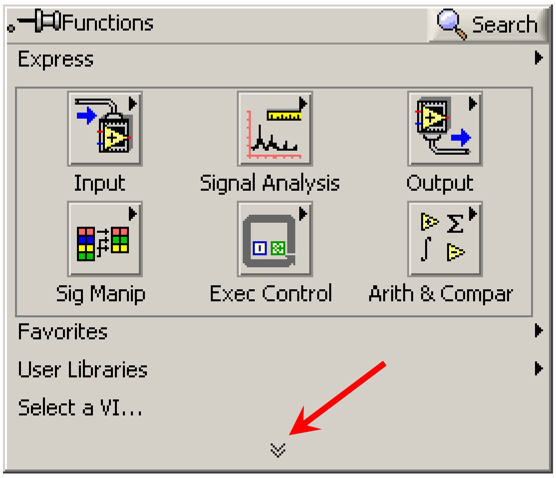

Figure 6-2, Functions Palette

Step 2

- **Expand** the Functions Palette by clicking the down-pointing chevrons as indicated by the red arrow in Figure 6-2 (click again to shrink the palette back to the Express view).
- Click on **Measurement I/O** sub-palette and then on **NI ELVISmx**. You will recognise the instruments you have been working with. This palette gives you access to functions to control the ELVIS hardware components from you program. Click on **NI ELVISmx Function Generator** and drop it on the Block Diagram.

The **Function Generator Configuration** dialog box appears when you place the Express VI on the block diagram. When you place an Express VI on the block diagram, the configuration dialog box for that VI always appears automatically.

- Set the controls as per Figure 6-3.

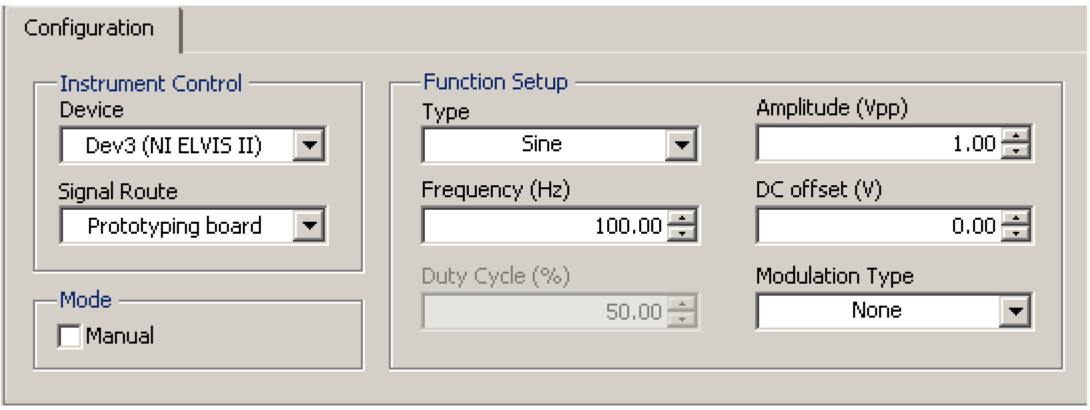

Figure 6-3

- Make sure the control Device is set to **NI ELVIS II;** the actual device number may differ from one workstation to another.
- Save and close the Configuration dialog box by clicking OK.

The VI you just placed on the Block Diagram is in essence a function that drives the ELVIS DAC in a fashion such that it behaves like a function generator. The FGEN instrument you have been using so far makes use of the save function.

- Take a look at the blue element that has just appeared on the Block Diagram (Figure 6-4). This is an Express VI. On the left of the VI are the function's input parameters and on the right side the returned values. Move the mouse over the coloured arrows on both sides of the VI to show the name of the input and output parameters.

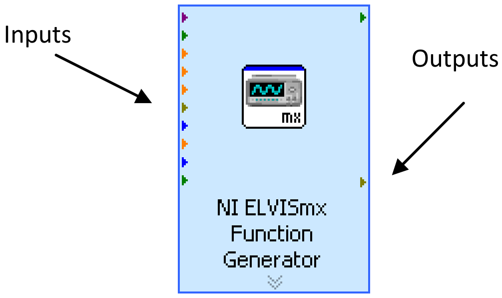

Figure 6-4, Function Generator VI

Step 3

We have just added to our program the ability to generate a waveform via the ELVIS Function generator. Now we want to add the ability to read then Pre Amp output waveform and make it available inside the program for further processing. For this purpose we can use the ELVIS oscilloscope.

- On the Functions palette, click on **NI ELVISmx Oscilloscope** and drop it on the Block Diagram. The **Oscilloscope Configuration** dialog box appears. Configure it as shown in Figure 6-5 (same settings for both Channel 0 and Channel 1) and then close the Configuration dialog box.

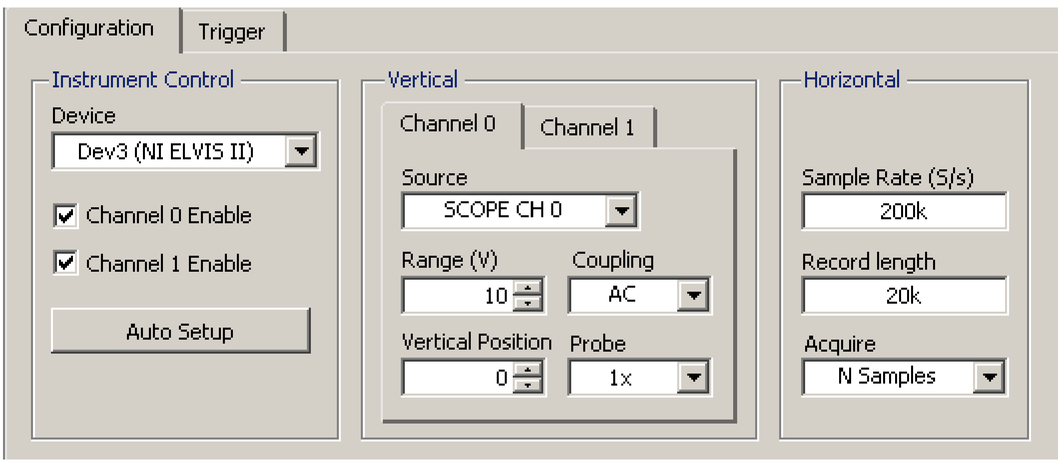

Figure 6-5

<u>Note the values in the Source control: you are instructing Oscilloscope VI to read data from "SCOPE CH0" and "SCOPE CH1", which are the two BNC on the left-hand side of the ELVIS device.</u>

➢ Take a look at the inputs and outputs available on the new VI just appeared on the Block Diagram.

Step 4

➢ Close the Measurement I/O palette by clicking on it and go back to the Express palette.

Step 5

Re-arrange the Block Diagram as shown in Figure 6-6.

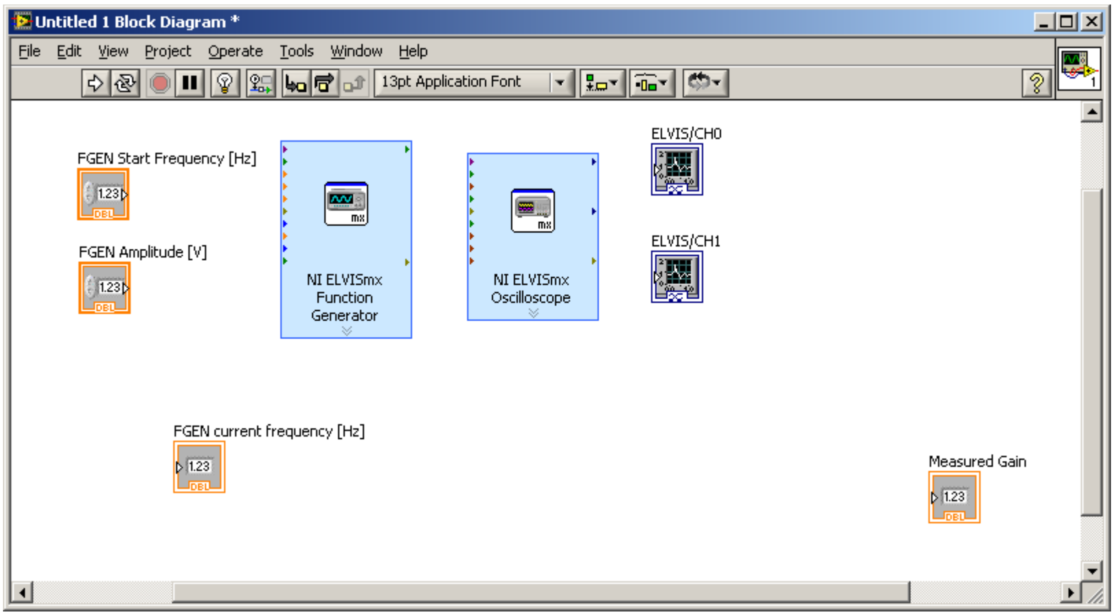

Figure 6-6

Step 6

Now let's wire the various components together.

➢ Move the cursor over the **arrow** on the **FGEN Start Frequency [Hz]** control terminal (Figure 6-7). The cursor becomes a wire spool which is the **Wiring tool**.

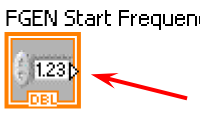

Figure 6-7

➢ When the Wiring tool appears, click the **arrow** on the **FGEN Start Frequency [Hz]** control terminal and then click the **arrow** on the **Frequency (Hz)** input of the **NI ELVISmx Function Generator VI** to wire the two objects together.

A wire appears and connects the two objects.

Data flows along this wire from the **FGEN Start Frequency [Hz]** control terminal to the NI ELVISmx Function Generator VI.

Step 7

In a similar manner, connect (see Figure 6-8):

- **FGEN Amplitude p-p [V] control** terminal to the **Amplitude (Vpp)** input of the NI ELVISmx Function Generator;
- The **error out** output of ELVISmx Function Generator to the **error in** input of ELVISmx Oscilloscope;
- The **ELVIS/Channel 0** output of ELVISmx Oscilloscope to the **ELVIS/CH0** graph indicator terminal;
- The **ELVIS/Channel 1** output of ELVISmx Oscilloscope to the **ELVIS/CH1** graph indicator terminal;
- The **FGEN Start Frequency [Hz]** control terminal to the **FGEN current frequency [Hz]** indicator terminal;

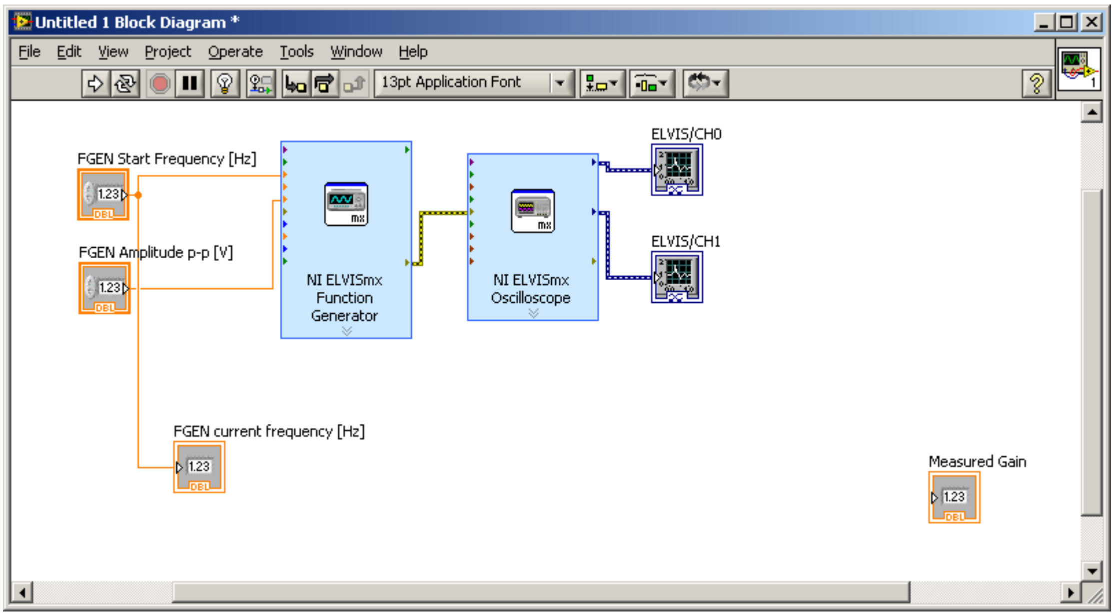

Figure 6-8

Step 8

- Select **File » Save** to save the VI.

7. Running the VI

Step 1

We want the function generator to produce a 50Hz sine wave with 0.4Vpp amplitude.

- On the Front Panel set the VI controls to:
  FGEN Start Frequency [Hz] = 50
  FGEN Amplitude p-p [V] = 0.4

Step 2

- Run the VI.

When you run the VI, the values you set in the Front Panel controls will be passed to the Function Generator, which in turn will invoke the underlying function to drive the ELVIS DAC to produce the desired waveform.

Similarly, once the Function Generator has been set up, the VI will return an error message that will travel along the wire on to the Oscilloscope VI. Assuming that the error message is "no error" the Oscilloscope VI will invoke the underlying function and perform a data acquisition according to the settings previously saved in the Oscilloscope configuration dialogue box.

After the measurement has taken place, the acquired data will travel along the wires from the Oscilloscope VI output terminals to the ELVIS/CH0 and ELVIS/CH1graphs controls and will be displayed on the Front Panel.

You should now have two sine waves in the two Front panel's graphs. One should be 10 times the amplitude of the other, 10 being the nominal gain of your Pre Amp by design.

The **Measured Gain** indicator will read 0, since it is not connected to anything.

Step 3

So far, we have written a piece of code that generates a sine wave and acquires data from two analogue channels (CH0 and CH1) of the ELVIS device. Now we want to make some calculation on those data and work out the Gain of the Pre Amp.

- Go back to the Block Diagram. On the Functions Palette, under Express, select the **Signal Analysis** subpalette and then the **Amplitude and Level Measurement VI**. Drop it on the Block Diagram. When the Configuration dialog box opens, tick **RMS**: we want to calculate the RMS value of the acquired signal. Other measurements are available but they are of no interest to us right now.

Step 4

- Repeat step 3 and add another **Amplitude and Level Measurement VI** configured to calculate **RMS**.

Step 5

- On the Express Palette, now select **Arithmetic and Comparison** and then **Express Numeric**. Drop a **Divide** function on the Block Diagram.

Step 6

Re-arrange the various terminals and wire them together as shown in Figure 7-1. You have to:

- wire Oscilloscope output **Channel 0** to the **Signals** input of one of the Amplitude and Level Measurement VIs and output Channel 1 to the other.
- wire the two Amplitude and Level Measurement VIs **RMS** outputs to the **Divide** function so that to divide the Pre Amp output voltage RMS (CH1) by the Pre Amp input RMS (CH0) and obtain the Pre Amp Gain.
- wire the **Divide** output to the **Measured Gain** indicator.

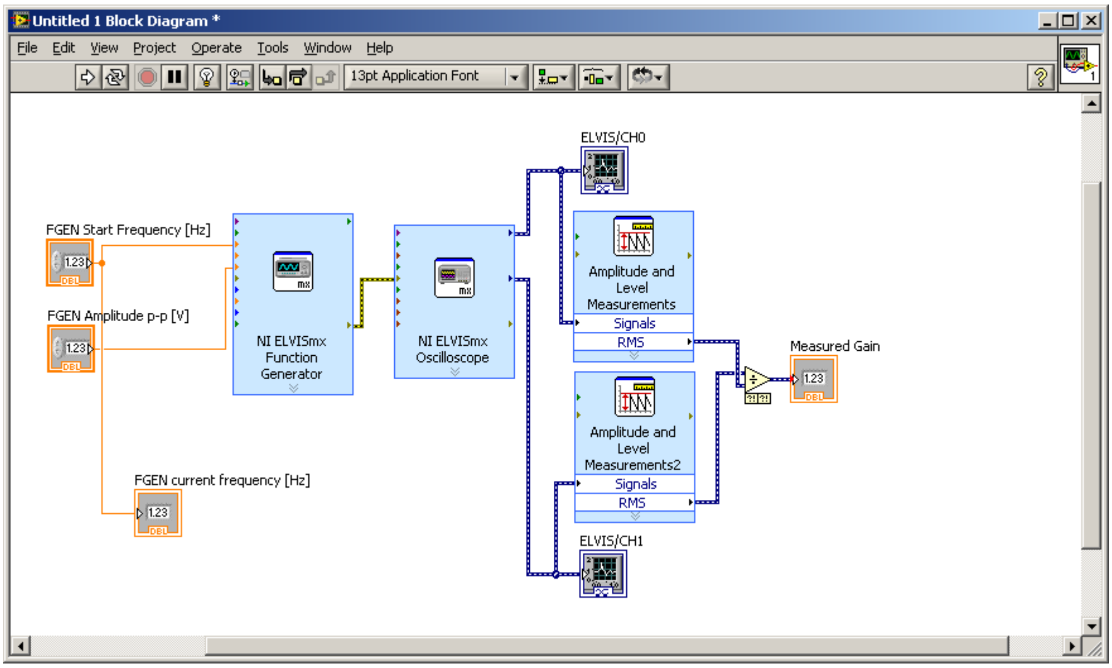

Figure 7-1

Step 7

- Run the VI.

The Gain indicator will now display the measured gain of the Pre Amp which should be about 10.

Step 8

Now we want to measure the Gain when the Pre Amp input signal frequency is 20kHz and 60kHz.

- Set the value of control **FGEN Start Frequency [Hz]** accordingly and run the VI.
- What is the Gain?

Tip: right click on the Graph control and select **Visible Items, Scale Legend** and then **Graph Palette** to show some useful tools for controlling the graph's behaviour. You may need to re-arrange the items on the Front Panel.

8. Adding the Loop

Your program, so far, allows you to measure the Pre Amp's Gain for a certain frequency which you have to set manually on the VI's Front panel. If you want to plot the Gain curve from 1kHz to 100kHz in 1kHz step you will have to set up and run the VI 100 times. This is obviously tedious, time consuming and prone to errors by the operator. Let's automate this process by making use of a "loop".

Step 1

- Go back to the Block Diagram. On the Functions Palette, under Express, select the **Execution Control** and then click **While Loop**. Drop it in the Block Diagram on the

top-left corner and drag it to the **bottom-right corner** to include all the elements already on the Block Diagram (use <Ctrl-Z> to undo, if you make a mistake).

- Click on the **Stop** terminal just appeared on the Block Diagram and delete it using the **Del** key.
- Right click anywhere on the **While Loop**'s grey border and select **Replace with For Loop.**
- Add a **Multiply** and an **Add** functions from the **Express Numeric** subpalette inside the For Loop.

Step 2

Go back to the Front Panel and add

- a Numeric Control and label it "Number of Steps";
- a **Numeric Control** and label it **"FGEN Step Frequency [Hz]";**
- an **XY Graph** and label it **"Gain Plot";** change the X-axes label to "Frequency [Hz]" and the Y-axes label to "Gain".
- Re-arrange the elements as necessary. You should end up with something like Figure 8-1.

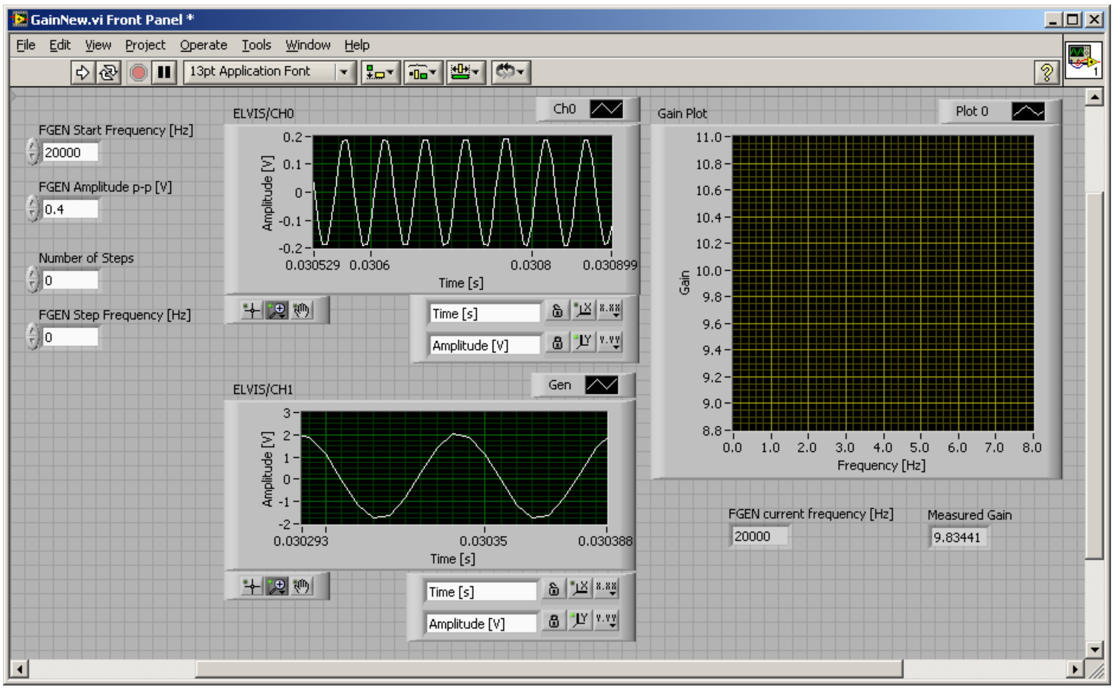

Figure 8-1

Step 3

Let's wire the controls and indicators just added to the VI. Go back to the Block Diagram.

- Re-arrange the blocks as necessary and wire them together as shown in Figure 8-2.

Tip: to delete a wire, click on it and press the **Del** key.

Tip: to get rid of broken wires use <Ctrl-B>.

- Double click on the Build X-Y Graph VI and uncheck Clear data on each call

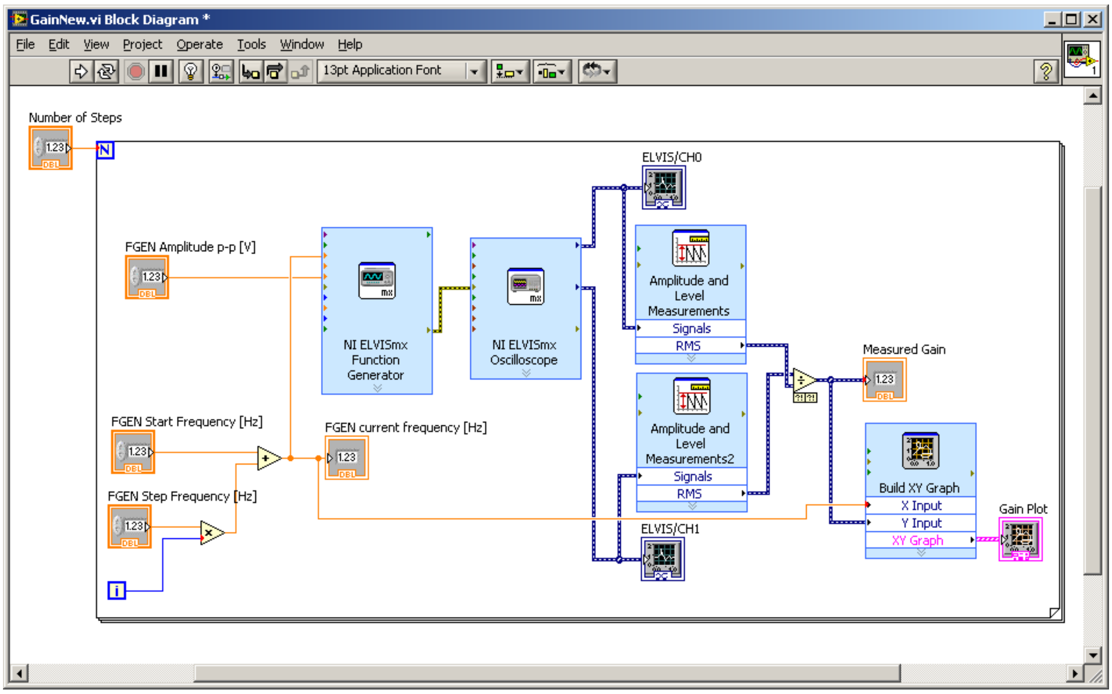

Figure 8-2

Step 4

- Select **File»Save** to save the VI.

Step 5

- Set the Front Panel controls to:
  FGEN Start Frequency [Hz] = 1000
  FGEN Amplitude p-p [V] = 0.4
  Number of Steps = 50
  FGEN Step Frequency [Hz] = 1000
- Run the VI.

While the VI is running pay attention to the values of displayed by the various indicators.

- Do you understand what's happening?

When the VI runs, the code inside the For Loop is executed a number of time equal the value of the For Loop's **N** input; in our case, the Loop's **N** input is wired to the control **Number of Steps** which means the For Loop will execute 50 time.
The **i** inside the For Loop is the loop's **index**. Its value is incremented by 1 (starting from 0) each time the For Loop executes.
the **Add** and **Multiply** functions have been wired to the controls and indicators so that the following algorithm is implemented:

FGEN current frequency [Hz] = FGEN Start Frequency [Hz] + FGEN Step Frequency [Hz] * i

where **i** = 0, 1, …, **Number of Steps** -1

Therefore **FGEN current frequency [Hz] =** 1000, 2000, 3000, …, 50000

At each iteration of the For Loop, the program will (in this order):

1. calculate the frequency **FGEN current frequency [Hz]**

2. generate the sine wave using the ELVIS hardware DAC
3. acquire 2 waveforms (channel 0 and channel 1) using ELVIS ADC
4. plot the two acquired waveforms
5. compute the RMS of the acquired waveforms
6. calculate the Gain dividing the CH1 RMS by CH0 RMS
7. plot the Gain

9. Remarks

1. In classical text programming, the data flow is dictated by the sequence in which the program statements are written: execution starts at the top of the text file and proceeds towards the bottom.
2. In LabVIEW, the order of execution is dictated by the **availability of data** at the VIs input terminals, not by the position of the Vis on the Block Diagram. For instance, the Divide functions will not execute until data is available at both of its input terminals i.e. until the two RMS values are calculated. In turn the RMS value is not calculated until the waveform data is returned by the Oscilloscope and so forth.
3. LabVIEW programs are usually written from the left of the screen to the right, but this is just a convention. You can place your VIs and terminals anywhere on the Block Diagram. The order of execution will not change, since it is dictated the by flow of the data along the wires, not their relative position.

10. Additional work

If you have the time, expand the Functions palette (click on the down-pointing chevrons) and take a look at the vast array of functions provided by LabVIEW. There is a whole host of analytical functions under the **Mathematics** and **Signal Processing** palettes, and a lot of functions to drive instruments under **Instrument I/O**. In addition to the functions you find in the basis package (the one you are using right now) there are several LabVIEW add-ons to fulfil special tasks ranging from motion control, to vibration analysis to manufacturing plant's control.